



PowerFit

Gym Management system

Project Documentation

Executive Summary

The **Maharaja Gym Management & Retail POS System** is an all-in-one desktop application designed to streamline fitness center operations, membership tracking, workout scheduling, and storefront retail point-of-sale (POS) operations. Built natively using **C# .NET Windows Forms** and **Microsoft SQL Server**, the system eliminates manual administrative overhead, safeguards corporate financial health with analytical reporting, and ensures cross-device portable pathing for storage security.

System Architecture & Tech Stack

- **Frontend Framework:** .NET Framework C# Windows Forms (WinForms)
- **Database Engine:** Microsoft SQL Server (MSSQL Relational DBMS)
- **Data Access Layer:** ADO.NET (`SqlConnection`, `SqlCommand`, `SqlDataReader`, `SqlDataAdapter`)
- **Architecture Pattern:** Layered Component Architecture with a Global Runtime Context State (`UserSession`)

Database Schema Blueprint

The system relies on a highly integrated, relational database schema optimized using foreign key constraints and transactional consistency rules. .

Users Table (**users**)

Tracks all physical individuals interacting with the infrastructure. High-level permissions are split using numeric tracking tags.

- **id** (VARCHAR(20), Primary Key) — Formatted dynamically: Admin (**ADM...**), Staff (**STF...**), Coach (**COA...**), Member (**MEM...**).
- **name** (VARCHAR(100))
- **email** (VARCHAR(100), Unique)
- **password** (VARCHAR(255)) — Secured by UI Masked Input system password structures.
- **role** (INT) — 1: Admin, 2: Cashier/Staff, 3: Coach, 4: Member.
- **phone** (VARCHAR(20))
- **address** (TEXT)
- **salary** (DECIMAL(18,2))
- **status** (INT) — 1: Active, 0: Suspended/Inactive.
- **dob** (DATE)
- **gender** (VARCHAR(10))

Gym Plans Table (**gym_plans**)

Stores structural membership access levels.

- **id** (VARCHAR(20), Primary Key) — (e.g., **PLAN001**, **PLAN002**)
- **name** (VARCHAR(50))
- **description** (TEXT)
- **price** (DECIMAL(18,2))
- **duration_months** (INT)
- **access_level** (VARCHAR(20))

Memberships Table (**memberships**)

Maintains precise, extendable timelines for customer privileges.

- **id** (VARCHAR(50), Primary Key) — GUID string tokens.
- **user_id** (VARCHAR(20), Foreign Key → **users.id**)
- **plan_id** (VARCHAR(20), Foreign Key → **gym_plans.id**)
- **start_date** (DATE)
- **end_date** (DATE) — Protected by a non-retroactive renewal timeline algorithm.
- **payment_status** (INT)

Store Items Table (**store_items**)

Tracks physical inventory supplements and corporate asset listings.

- **id** (VARCHAR(20), Primary Key) — (e.g., **ITM...**)
- **name** (VARCHAR(100))
- **description** (TEXT)
- **price** (DECIMAL(18,2))
- **stock** (INT)
- **category** (VARCHAR(50)) — **GymItem** or **Memberships**.
- **expiry_date** (DATE, Nullable)
- **image_path** (VARCHAR(500)) — Saved via localized sandboxed relative directories.
- **min_stock_level** (INT)

Schedules & Exercises

- **schedules**: Connects trainers to clients (**id**, **title**, **member_id**, **coach_id**, **days_per_week**).
- **schedule_exercises**: Data junction map connecting workouts step-by-step (**schedule_id**, **exercise_id**, **sets**, **reps**, **day_of_week**, **notes**).

Core Core Feature Specifications

Global State Management (**UserSession**)

To prevent passing complex structural entities between screens, the platform relies on a memory-cached object tracker. When a user authenticates through **loginForm**, their credentials map onto a static instance globally visible to all running threads:

- Allows the workout creation modules to seamlessly auto-populate **@coach_id** with whoever is currently logged in, allowing both **Admins** and **Coaches** to draft schedules without explicit code hardcoding.
- Enables **Self-Profile Redirection**. When clicking settings, the registration form boots into a protective layout lock that blocks the editing of privileges (**cbRole.Enabled = false**) while opening updating operations specifically for **UserSession.UserID**.

POS Membership Renewal Engine

The checkout counter runs a complex calculation rule when handling membership plans:

- **If Expired (**Expiry < Now**)**: The system completely truncates the past, resetting the renewal to accumulate exactly starting from **Today** (**DateTime.Now.AddMonths(duration)**).

- **If Actively Running (Expiry >= Now):** The engine respects their current access timeline, seamlessly appending the purchased period onto their **future expiration date**.

Sandboxed Asset System

To guarantee that image paths do not break when migrating the app between multiple development workstations, the product creation workspace utilizes an automatic asset cloner. When browsing an image, the file stream copies the graphic straight into a local sandbox:

`AppDomain.CurrentDomain.BaseDirectory \ Images \` The relative installation path is saved to the database, rendering images securely using non-locking `FileStream` reads to prevent memory locking.

Real-Time Status Filters

The management grid features dynamic check filtering logic. When ticking the **Expired Items** filter, the SQL query intercepts runtime variables and injects chronological parameter exclusions (`expiry_date < CAST(GETDATE() AS DATE)`), isolating out-of-date stock items instantly.

Native Administrative Reporting (No External Dependencies)

To avoid setup errors caused by missing DLL components on end-user machines, reporting is handled by the **Native .NET GDI+ Printing Canvas Engine** (`System.Drawing.Printing`). The system converts raw analytical tables (`DataTable`) directly into graphical layouts by manually mapping x/y scaling metrics onto portrait bounds, supporting direct **Print-To-PDF** execution out-of-the-box.

Security & Validation

Controls

- **SQL Injection Guardrails:** Every database connection strictly enforces strongly-typed parameters (`SqlCommand.Parameters.AddWithValue`), neutralizing malicious SQL injections.
- **System UI Input Validation:** All forms utilize interceptors (`checkForm()`) that check constraints via `string.IsNullOrEmpty` prior to invoking connection state buffers.
- **Password Privacy Mask:** The login panel includes an interactive PictureBox toggle modifying the character encoding structure natively (`txtPassword.UseSystemPasswordChar`) alongside runtime resource changes matching visual feedback expectations.

Implementation & Operational Guidelines

6.1. Deployment Requirements

- **IDE:** Microsoft Visual Studio (2022 or newer recommended) with the **.NET Desktop Development** workload installed.
- **Database Management:** Microsoft SQL Server Management Studio (SSMS) or SQL Server Express LocalDB instance.

Step-by-Step Deployment Protocol

Step 1: Restore the Database Backend Engine

1. Navigate directly into the local project directory structure and open the folder named: **[Project_Root]\db Script**
2. Locate the master schema script file: **script.sql**.
3. Open **SQL Server Management Studio (SSMS)**, connect to your primary SQL Server target instance, and open this script file.
4. Execute (**F5**) the entire script. This operation will programmatically build the database containers, map out the relational schemas, and insert initial seeding data configurations (roles, statuses, and fallback user profiles).
- 5.

Step 2: Configure the Target Application Connection String

1. Open the project solution workspace (`.sln`) inside Visual Studio.
2. Locate and open the global `dbConnect.cs` helper class container.
3. Update the data source attributes inside your connection string to explicitly mirror your live production server context environment:

C#

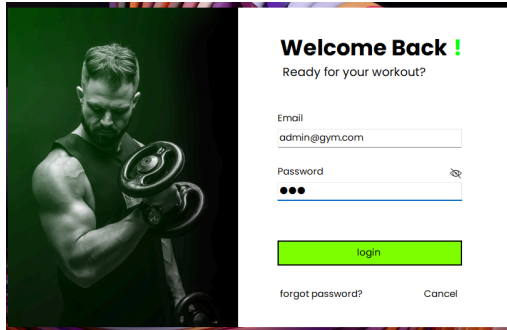
```
public string connection()
{
    return @"Data Source=YOUR_SERVER_NAME;Initial
Catalog=Gym_Management_System;Integrated Security=True;";
}
```

Step 3: Compile and Distribute Standalone Executables

1. Shift your Visual Studio solution build configuration toggle from **Debug** to **Release**.
2. Right-click the primary workspace name in the Solution Explorer panel and choose **Clean Solution**, followed immediately by **Build Solution**.
3. Once compiled successfully, navigate to your local workspace output terminal subdirectory: `[Project_Root]\bin\Release\`
4. Copy this compiled standalone asset package. It contains your compiled binary files, configuration schemas, and sandboxed asset modules. It can be instantly deployed onto any client Windows terminal desktop without requiring an IDE!

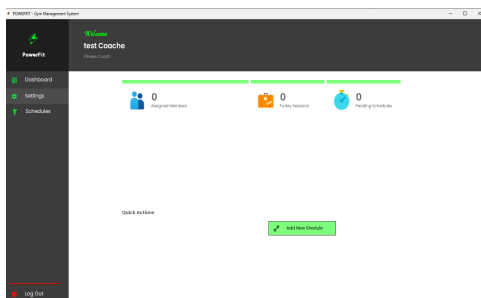
ScreenShots

Login Form

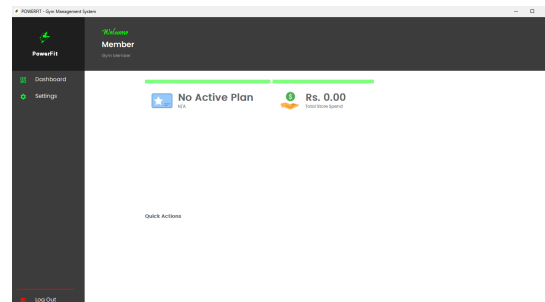


The login form is titled "Welcome Back !" and asks "Ready for your workout?". It features a background image of a man lifting a dumbbell. The form includes fields for "Email" (pre-filled with "admin@gym.com") and "Password" (masked with dots). A red "login" button is positioned below the password field. At the bottom, there are links for "forgot password?" and "Cancel".

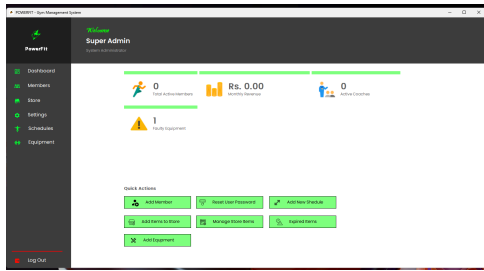
- **Role-Based Dashboards:** The system dynamically adjusts the interface based on the user's role (Admin, Coach, or Member).
 - **Admin:** Oversees members, inventory, gym equipment, and financial reporting.
 - **Coach:** Manages assigned members and session schedules.
 - **Member:** View their active plan and total spending.



Dashboard (Coach)

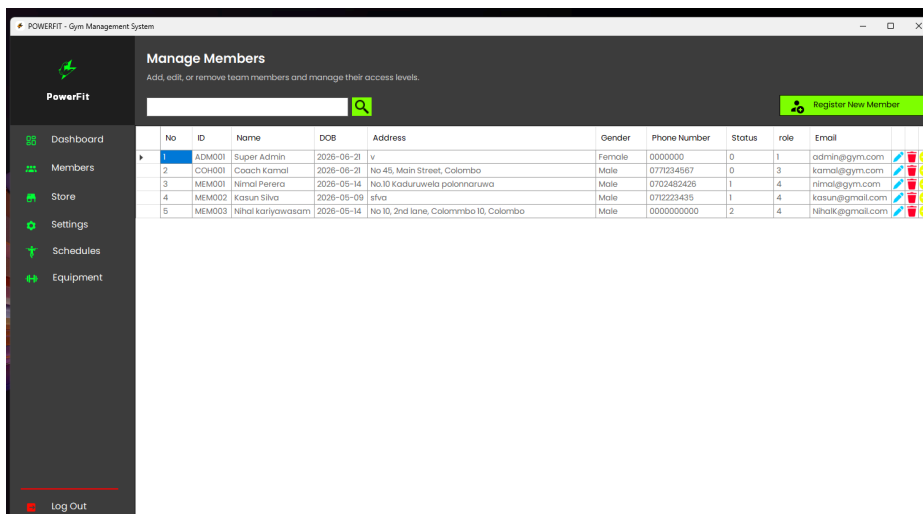


Dashboard (Member)



Dashboard (Admin)

- **User & Membership Management:**
 - Registers and tracks individuals (Admins, Staff, Coaches, Members).
 - Handles membership renewals with a specific algorithm: it either resets start dates upon expiry or extends the expiration date if the membership is still active.¹



Add Member and Update Member form

The screenshot shows a web browser window titled "New User Registration" with a close button. The header features the "PowerFit" logo and the title "New User Registration" with the instruction "Enter the hardware specifications." The form fields are as follows:

- Name:** A single-line text input field.
- Phone Number:** A single-line text input field.
- DOB:** A date picker showing "Sunday, June 2" with a dropdown arrow.
- Role:** A dropdown menu.
- Gender:** A dropdown menu.
- eMail:** A single-line text input field.
- Address:** A multi-line text area.
- Status:** A dropdown menu.
- Buttons:** A green "Save" button and a light green "Cancel" button.

POS & Retail Inventory:

- Includes a "Gym Retail Shop" form for managing product stock levels and processing customer checkouts.¹
- Allows filtering for expired stock to maintain inventory health.

Gym Retail Shop Form

POWERFIT - Gym Management System

Gym Retail Shop

Track product stock levels, manage sales, and update retail items for gym members.

Store Items | Store Settings | Hold Orders

Basic Monthly

3,500.00

Standard Monthly

5,500.00

Student Special

3,000.00

Pro Quarterly

15,000.00

Annual Elite

50,000.00

VIP Family

85,000.00

Current Order

Gym Gloves (L)	2,500.00	2	5,000.00
Creatine Monohydrate	4,500.00	1	4,500.00
Personal Training 10S	15,000.00	1	15,000.00
Annual Elite	50,000.00	1	50,000.00
Pro Quarterly	15,000.00	1	15,000.00
Total			89,500.00

Log Out

formAddStoreItem

Add New Item to Store

Enter the hardware specifications.

Creatine Monohydrate

Name : Pure creatine for strength and power

Category : 2026-06-30

Price : 15

Stock : Gymitem Min Stock Level : System.Drawin

Expiry Date : Sunday , June 21, 2026

Description : 4500.00

Image : C:\Users\User\Pictures\Screenshot 2026- Choose image

Update Cancel

Manage Store Items

Store Items

Input asset details, serial numbers, and physical conditions to update the facility registry.

Add New Item

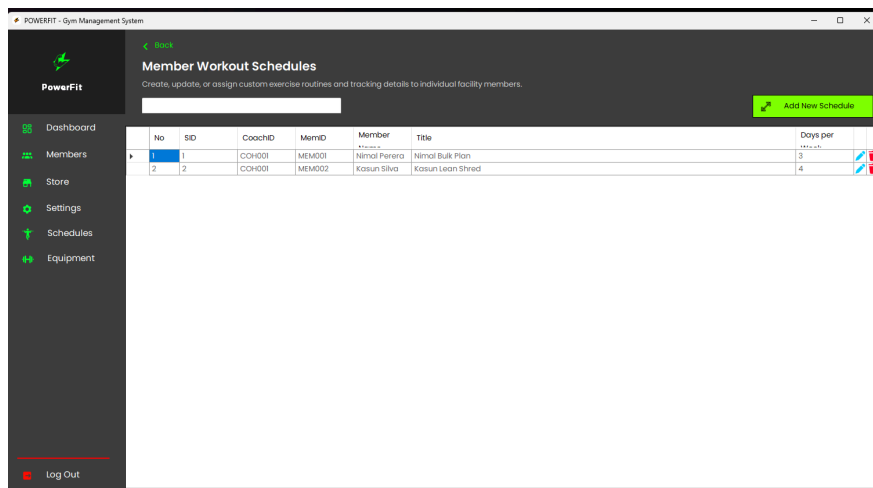
☐ Expired Items

ID	Name	Description	Price	Stock	Category	Expiry Date	Min Stock	
ITM001	Whey Protein 2kg	High-quality whey protein for muscle recovery	18500.00	20	Gymitem	2027-12-31	5	
ITM002	Creatine Monohydrate	Pure creatine for strength and power	4500.00	15	Gymitem	2026-06-30	3	
ITM003	Pre-Workout Blast	Explosive energy for your training	8900.00	10	Gymitem	2026-10-15	2	
ITM004	Gym Gloves (L)	Premium grip gym gloves	2500.00	50	Gymitem	N/A	10	
ITM005	Monthly Pass	Standard gym access for 30 days	5000.00	999	Memberships	N/A	0	
ITM006	VIP Annual Membership	Unlimited access + Personal Trainer	45000.00	100	Memberships	N/A	0	
ITM007	Personal Training 10S	10 sessions with a certified coach	15000.00	999	Memberships	N/A	0	

Workout & Schedule Tracking:

- Enables coaches to create and modify custom workout schedules, mapping exercises to specific members and days of the week.

Manage the Members Workouts



Add Workout plans using this form

Modify Workout Schedule
Update active routine titles, modify weekly frequencies, or adjust assigned exercises and training notes

Coach : C000
Saman Kumara

Member : MEM003 - Nihal kariyawasam

Schedule Title : temp

Days Per Week : 3

Exercise : [+ Add](#)

Sets : Reps :

Day of Week :

Notes : [+ Add](#) [Clear](#)

All Exercises

ID	Exercise	Sets	Reps	Day of Week	Notes
3	Lat Pulldown	1	1	Day 02	temp

[Save](#) [Cancel](#)

Asset & Equipment Maintenance:

- Allows administrators to register and track gym equipment (like the "ProRun 9000" treadmill in your screenshots), including serial numbers, purchase dates, and warranty information.

Manage Gym Equipments

POWERFIT - Gym Management System

Back

Asset and Equipment

Input asset details, serial numbers, and physical conditions to update the facility registry.

+

Add New Asset

Dashboard

Members

Store

Settings

Schedules

Equipment

Log Out

ID	Name	Brand	Category	Purchase Date	Status	Model	Serial	Purchase Price	Warranty Expiry Date	
EQ-001	ProRun 9000	LifeFitness	Treadmill	1/15/2024 12:00:00...	Active	LF-TR-9000	SN-TR849201-X	4500.00	1/15/2026 12:00:00 AM	
EQ-002	Olympic Flat Bench	Hammer Str...	Bench Press	3/10/2024 12:00:00...	Under Repair	HS-BP-01	SN-BP10293...	850.00	3/10/2027 12:00:00 AM	
EQ-003	Hex Dumbbell Set 5-50lbs	Rogue Fitness	Dumbbell	6/1/2025 12:00:00...	Active	RF-HEX-550	SN-08550-Q	1200.00	6/1/2028 12:00:00 AM	
EQ-005	test 2	test 2	Dumbbell	6/20/2026 12:00:0...	New	test 2	test 2	100.00	6/20/2028 12:00:00 AM	

formAddAssets

Asset and Equipment Registration

Enter the hardware specifications.

Name :

ProRun 9000

Category :

Treadmill

Status :

Active

Brand :

LifeFitness

Model Number :

LF-TR-9000

Serial Number :

SN-TR849201-X

Purchase Date :

Monday , January 15, 2024

Purchase Price :

4500.00

Warranty Expiry Date :

Thursday , January 15, 2026

Update

Cancel

formAddAssets

Asset and Equipment Registration

Enter the hardware specifications.

Name :

Category :

Status :

Brand :

Model Number :

Serial Number :

Purchase Date :

Sunday , June 21, 2026

Purchase Price :

Warranty Expiry Date :

Sunday , June 21, 2026

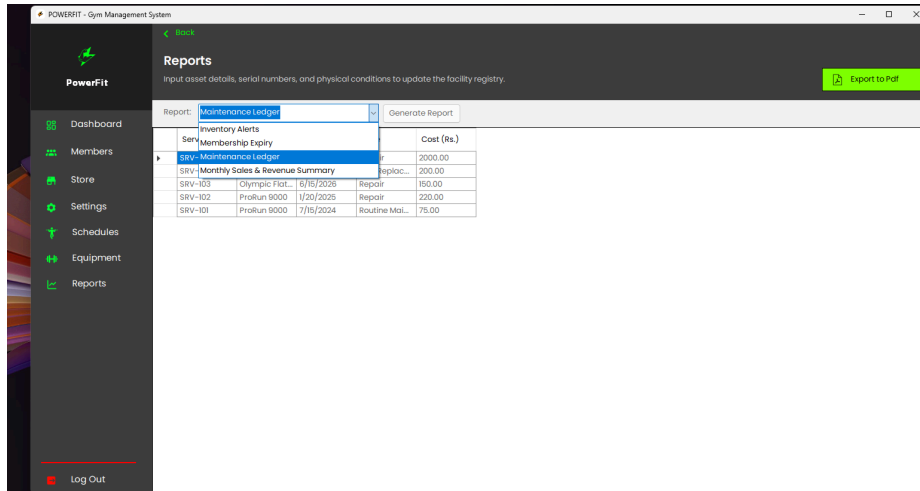
Save

Cancel

Native Administrative Reporting

- Comprehensive Data Insights:** Provides real-time visibility into key performance indicators, including "Maintenance Ledger" for equipment upkeep, "Inventory Alerts" for stock management, and "Membership Expiry" for subscription tracking.

- **Financial & Analytical Intelligence:** Generates "Monthly Sales & Revenue Summary" reports to support informed decision-making regarding the gym's fiscal health and profitability.



Group Members

Member	Form File Name
EC/2022/014 J P N N Senadheera	formReports.cs, formAddStoreItem.cs, formStore.cs , formSchedule.cs, formAddSchedule.cs, formSchedule.cs, formEquipment.cs,
EC/2022/033 M A P L Munashingha	MembersForm.cs, userRegistrationForm.cs, formManageStoreitems.cs, formPasswordReset.cs, FormAddExercise.cs, formChangePassword.cs formAddService.cs,
EC/2022/136 D I Udayanga	mainFrame.cs, adminDashboardForm.cs, loginForm.cs, settingsForm.cs , formAssetsInfo.cs, formAddAssets.cs, splashForm.cs